

2. What is flat file system? Explain the advantages of using DBMS over flat file system.
3. How logical data independence is different from physical data independence? Explain three level database architecture.
4. How primary key is different from super key? Explain composite key and partial key.
5. What are selection and projection operators in relational algebra? Given a table with schema

Student(Rollno(PK), Fname, Lname, Address, Level, Did(FK)),

Department(Did(PK), Dname, HOD)

Write an RA query to find the name of students who are in 'Bachelor' level living in 'Birgunj' and studies under the department with department

head 'Ramesh Sharma'.

6. How parameters can be passed to a stored procedure? Explain with an example.
7. How timestamp based algorithm uses timestamp to serialize the execution of concurrent transactions? Explain in detail.
8. What are different types of indexing used in database? How index can be created using SQL? Write with example.

2. Define database system. Write the merits and demerits of DBMS.

3. Draw and explain three schema architecture of the database system.

4. What are an entity and its attributes? Write the difference between a strong and weak entity set with an example.

5. Why integrity constraints are used in databases? Explain different types of integrity constraints with examples.

6. What is functional dependency? Explain its types with examples.

7. Differentiate between discretionary access control and mandatory access control with example.

8. Write the ACID properties of transactions. Explain the concepts of serializability in database transactions.

2. Define database and DBMS. Explain the benefits of DBMS in detail.

3. What are different SQL constraints to specify the rules for the data in the database? Explain.
 4. What is relational algebra? Explain different RA operations.
 5. How candidate key is different from primary key? Also explain foreign key in detail.
 6. What is a stored procedure? How can it be used in a database? Explain with an example.
 7. What are the major purposes of concurrency control? Explain two phase locking protocol in detail.
 8. What is database security? Describe steps of query processing.
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2. What is DBMS? Describe the advantages and disadvantages of using DBMS.
3. Describe 3-schema architecture of database.

4. What are roles and responsibilities of DBA.
5. Why indexing is essential in database? Differentiate dense index from sparse index with suitable example.
6. What are the states of transaction model? Describe the ACID properties of transaction.
7. Describe the use of ER diagram. Draw an ER diagram for a university to design a database for it.

8. What is DDL and DML queries?

Consider the following relation.

Customer name	Loan number
X	Lo1
Y	Lo2

Relation borrower

Loan number	Branch name	Amount
Lo1	B1	5000
Lo2	B2	6000

Relation loan

Write a relational algebra expression to find all customer who taken loan from branch "B1".

2. What is database? Discuss the importance of DBMS.
3. Define union compatibility. Given following relations, show the resulting relation of Director $\cap$ Actor and Actor $\cup$ Director?

Director

Fname	Lname
Deepak	Giri
Bhuwan	KC
Dipendra	Khanel

Actor

First name	Last name
Deepak	Giri
Rajesh	Hamal
Dipendra	Khanel

4. How Order By and Group BY Having Clause in SQL are different? Illustrate with suitable examples.

5. Why normalization is needed? When any relation is said to be in 1NF and 2NF?

6. How cost of query is measured during query processing? Construct query expression tree for the relational algebra query:

TLFName LName, Address(ODName="Research" A Department Dnumber=Dno
(Employee * Works_on))

7. How wait-for-graph is constructed? How can you use it to detect deadlock in a schedule of transactions? Support your answer with an example.

8. How read and write operations are performed in a transaction? List the various states where transactions can enter into.

2. What is DBMS? Describe the merits of using DBMS.

3. What is Data Model? What is the difference between hierarchical and network data model? what are its types?

4. Why indexing is essential in database? Differentiate dense index from sparse index with suitable example.

5. Discuss about materialized and pipelined evaluation of query expression in query optimization.

6. Explain with examples, how lost update and dirty read problems can occur in sentence of transactions?

7. How time stamp ordering protocol is used to ensure concurrency control?

8. What is the advantage and Disadvantage of using stored procedure? How can you create and execute stored procedure?

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